

Learning How Computers Work and How to Troubleshoot Them

The hands-on companion to the 32-lesson course. There is one lab per lesson, and each one takes you from understanding a part to repairing it with confidence.

HOW TO USE THIS GUIDE

This guide is written for someone learning alone, at their own pace, on machines they own. Each lab goes with the lesson of the same number. Watch the lesson, then do the lab, usually on the same day. Labs build on each other. Lab 1 starts a **machine record** and Lab 4 starts a **troubleshooting log**, and both are used through to the last lab. Keep them in one notebook or one text file.

The labs follow the course's six parts: Foundations (1–5), Hardware (6–13), the Operating System and Software (14–20), Networks and Security (21–24), Troubleshooting in Practice (25–29), and Building, Helping and Growing (30–32). The first labs mostly have you look and record. From Lab 25 on, you do full diagnoses.

Every lab has four parts

OBJECTIVES	What you will be able to do when you finish. Use them to check yourself.
MATERIALS	What to gather before you start. Tick each item off.
PROCEDURE	Lettered sections (A, B, C...) with numbered steps (A.1, A.2...). Do them in order.
TROUBLESHOOTING	What you are likely to see if a step doesn't go as expected, and what to try first.

Reading a step

When a step works differently on each operating system, it includes a block with one row per system. Do only the row for your machine. Text in monospace is either a command to type or a menu path, with › between each menu level.

WIN	Windows 10/11. PowerShell unless it says otherwise. Run: means press Win+R and type the text.
MAC	macOS 13 or later. Commands go in Terminal.
LINUX	Ubuntu, Debian or Fedora family. Commands go in any terminal. Package names are Debian-style.

CAUTION An orange box marks a step that can lose data, damage hardware or hurt you. Read it before you start the step, not after.

The method you will use in every lab

Lab 4 introduces the method and Lab 25 covers it in full. Every troubleshooting step in this guide is one turn of this cycle:

01 DEFINE Write down exactly what fails, when, and how often.	02 GATHER Find out what changed, collect logs and readings, and reproduce the fault.	03 HYPOTHESE Rank the possible causes by how likely and how cheap to test they are.
04 TEST ONE CHANGE Change one thing only, and make it a change you can undo.	05 EVALUATE Check whether it fixed the fault. If not, undo it.	06 LOG & REPEAT Write down what you did, then go on to the next hypothesis.

Ground rules

- 1. Back up first.** Do this before any hardware work, driver change, firmware visit or reinstall. Lab 20 shows you how to prove a backup works.
- 2. Change one thing at a time.** If you change two things, you won't know which one did anything.
- 3. Use administrator rights sparingly.** Open an admin terminal or use sudo only for the command that needs it.
- 4. Leave the power supply and swollen batteries alone.** Replace them; never open them.
- 5. Know when to stop.** Stop and get professional help if you smell burning, see physical damage on a board, or a drive is clicking while it still holds data that has no backup.

What a Computer Actually Is

Map your own machine to input, processing, storage and output.

OBJECTIVES

- OBJ-1** Name every external port on a computer you own and state what connects to it.
- OBJ-2** Place each part in one of four roles: input, processing, storage or output.
- OBJ-3** Find the make, model and basic specifications from inside the operating system.
- OBJ-4** Start a machine record that later labs will add to.

MATERIALS & TOOLS

- ☐ A desktop or laptop you are allowed to inspect
- ☐ Flashlight or phone torch
- ☐ Phone camera
- ☐ Notebook or a text file for your machine record
- ☐ Manufacturer's spec page for your model (look it up by model number)

PROCEDURE

A Identify the machine

- A.1** Find the model number printed on the machine. Laptops usually have it on the bottom; desktops on the back or side panel. Write it at the top of your machine record.

- A.2** Get the same details from the operating system, then check they match the label.

WIN	Settings > System > About —or— run: msinfo32
MAC	> About This Mac > More Info
LINUX	hostnamectl ; sudo dmidecode -s system-product-name

- A.3** Write down the processor, installed memory, storage capacity and OS version. You will explain each of these numbers in Labs 2, 6, 7 and 8.

B Survey the outside

- B.1** With the machine powered off, photograph every side that has ports. Use the flashlight to read the small icons and labels next to each port.
- B.2** List each port and match it to its name: USB-A, USB-C/Thunderbolt, HDMI, DisplayPort, Ethernet (RJ-45), 3.5 mm audio, SD card, power input. For each one, write what you would plug into it.

- B.3** Mark each port with its role: input (keyboard, mic), output (display, speakers), both (USB, Ethernet), or power.

c Classify every part

- C.1** Make a four-column table with the headings Input, Processing, Storage and Output. Put every component and peripheral you have found into a column. Some go in two columns: a touchscreen is both input and output.
- C.2** Add a fifth row for firmware and write where the firmware lives on this machine. The manufacturer's spec page or the OS details from step A.2 will help.
- C.3** Open the manufacturer's spec page. Circle anything listed there that you have not accounted for, such as internal slots, the Wi-Fi card or the battery. The lessons in Part II open these up.

TROUBLESHOOTING

IF YOU SEE	TRY
A port has no visible label	Compare its shape to the manufacturer's spec page. Many USB-C ports only show a lightning bolt or a D symbol if they also carry Thunderbolt or DisplayPort.
The OS reports a different model than the sticker	The board or chassis may have been replaced during a repair. Trust the firmware report (dmidecode or msinfo32) for the board, and the sticker for the chassis.
You cannot find the storage capacity	Look under Storage in Settings (Windows or macOS), or run lsblk on Linux. Lab 8 covers why the number may look smaller than advertised.
Unsure if a port is input or output	Ask which way the data flows most of the time. USB and Ethernet carry data both ways, so list them as both.

Numbers, Bits, and How Machines Represent Everything

Convert between bases and explain why a drive looks smaller than the box says.

OBJECTIVES

- OBJ-1** Convert numbers between binary and decimal by hand.
- OBJ-2** Explain how a byte turns into a character through an encoding.
- OBJ-3** Compare a file's actual size with its size on disk.
- OBJ-4** Explain the gap between advertised and reported drive capacity.

MATERIALS & TOOLS

- ☐ Paper and pencil
- ☐ Any computer
- ☐ A small plain-text file (create one) and a photo file
- ☐ A calculator app with a Programmer mode (optional, for checking)

PROCEDURE

A Binary by hand

- A.1** Write the place values for eight bits: 128 64 32 16 8 4 2 1. Convert 00101101 to decimal by adding the place values that hold a 1. (Answer: 45.)
- A.2** Convert 200, 7 and 255 to binary. Start from the largest place value that fits, subtract it, and move right. Check your answers in Programmer mode.

WIN	Calculator > menu > Programmer
MAC	Calculator > View > Programmer
LINUX	GNOME Calculator > Mode > Programming <code>—or— echo "obase=2; 200" bc</code>

- A.3** Write down why 255 is the largest number a single byte can hold, and how many different values a byte can represent (256).

B Text is numbers under an agreement

- B.1** Create a file named hello.txt that contains exactly: Hi

- B.2 Look at its bytes. You should see 48 69, the ASCII codes for H and i, possibly followed by a line ending.

WIN	PowerShell: Format-Hex .\hello.txt
MAC	xxd hello.txt
LINUX	xxd hello.txt --or-- od -An -tx1 hello.txt

- B.3 Add an accented character (é) and save the file as UTF-8. Look at the bytes again: that one character now takes two bytes. This is why text opened with the wrong encoding shows up as garbage symbols.

C Real size versus reported size

- C.1 Check hello.txt's size in bytes and the space it actually takes on disk.

WIN	Right-click > Properties > Size vs Size on disk
MAC	Finder > Get Info (size vs on disk)
LINUX	stat -c '%s bytes, %b blocks' hello.txt ; du -h hello.txt

- C.2 Write down why a 3-byte file takes up 4 KB. The file system hands out space in whole blocks (clusters).
- C.3 Look at your drive's total capacity as the OS reports it. Divide the capacity on the box (in bytes) by 1,073,741,824 to get GiB. For example, a 1 TB drive is 1,000,000,000,000 bytes, which is about 931 GiB. Compare your result with what the OS shows.

TROUBLESHOOTING

IF YOU SEE	TRY
Your binary answer is off by one place value	Check that you wrote eight bits, and that the leftmost bit is worth 128, not 256.
Format-Hex or xxd shows extra bytes at the start	Those are a byte-order mark (EF BB BF for UTF-8). Some editors add one. It is harmless, but it can confuse scripts.
The line ending shows as 0D 0A or as 0A	0D 0A is the Windows line ending (CRLF); 0A is macOS and Linux (LF). This difference explains many files that look broken when opened on another system.
Size on disk shows 0	Very small files can be stored inside the file system's own records, so they use no separate block. This is normal on NTFS.

How the Parts Talk: A Working Mental Model

Narrate a real startup, one stage at a time.

OBJECTIVES

- OBJ-1** Draw the processor, memory and storage triangle and label how fast and how permanent each part is.
- OBJ-2** Watch a cold start and name each stage as it happens.
- OBJ-3** Tell the difference between memory filling up and storage filling up on a live system.

MATERIALS & TOOLS

- ☐ A computer that is fully shut down
- ☐ Paper for a timeline sketch
- ☐ A phone stopwatch

PROCEDURE

A Draw the model

- A.1** Draw three boxes, CPU, RAM and Storage, with arrows between them labelled bus. Next to each box write two words: one for speed (fastest, fast, slow) and one for permanence (volatile or non-volatile).
- A.2** Under the diagram, write one sentence that explains the difference between memory and storage without using the word space.

B Narrate a cold start

- B.1** Shut down fully, not sleep or hibernate, and wait ten seconds.

WIN	Hold Shift while clicking Shut down (turns off Fast Startup for this shutdown)
MAC	> Shut Down
LINUX	systemctl poweroff

- B.2** Start the stopwatch and press power. Say each stage out loud as you see it and note the time: fans or lights come on, then the logo appears (firmware and POST), then the spinner or boot menu (bootloader), then the login screen (OS loaded), then the desktop is ready (your programs starting).

- B.3 Draw your times as a timeline. Circle the longest stage. Lab 13 explains what is happening in each one.

c Watch memory and storage separately

- C.1 Open the system monitor and write down memory in use and free storage.

WIN	Task Manager (Ctrl+Shift+Esc) > Performance
MAC	Activity Monitor > Memory ; System Settings > General > Storage
LINUX	free -h ; df -h /

- C.2 Open five browser tabs and a large document. Write down memory and storage again. Memory should go up and storage should barely change. Close everything and watch memory come back down.

TROUBLESHOOTING

IF YOU SEE	TRY
The machine goes straight to the desktop with almost no firmware stage	Fast Startup or hibernation may be on, or the firmware may hide the logo. Try a restart instead of a shutdown, because a restart always does a full cold boot on Windows.
Memory barely drops after closing programs	The OS keeps recently used data cached in memory. That is normal. Cached memory is freed as soon as something else needs it.
You cannot tell where the bootloader stage ends	On most modern machines this stage lasts under a second. Write it down as a combined firmware-to-OS stage.

The Troubleshooting Mindset and Method

Write a full diagnostic plan before touching anything.

OBJECTIVES

- OBJ-1** Turn a vague complaint into a precise problem statement.
- OBJ-2** Write at least three ranked hypotheses for the fault, each with a test that proves or rules it out.
- OBJ-3** Build a plan where each step changes only one thing.
- OBJ-4** Start a troubleshooting log you will reuse for the rest of the course.

MATERIALS & TOOLS

- ☐ The fault scenario below
- ☐ Paper or a text file set up as a troubleshooting log

PROCEDURE

A The fault

- A.1** Read the scenario: "My laptop has been really slow since last week, and yesterday it shut itself off while I was on a video call. It's plugged in most of the time. I installed a new video app and my kid has been using it on the bed."

B Define and gather

- B.1** Rewrite the complaint as one precise sentence: what happens, when it happens, and how often. Keep what you observed separate from what you are guessing.
- B.2** List every question you would ask the owner. Include what changed, when it started, and whether it can be reproduced. Mark which questions the scenario already answers.
- B.3** List the information you could collect without changing anything, such as temperatures, the list of recent installs, and event or crash logs. Leave the commands blank for now; later labs fill them in.

C Hypothesise and plan

- C.1** Write at least three hypotheses, for example overheating from blocked vents, a runaway process from the new app, or a failing battery or charger. Rank them by how likely they are and how cheap they are to test.

- C.2 For each hypothesis, write one test that changes only one thing, and what result would confirm it or rule it out.
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- C.3 Write your stop conditions: when you would back up first, and when you would hand the machine to a professional.
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D Set up the log

- D.1 Make a log template with these columns: Date · Change made · Result · Next step. Keep it for every fault you work on from now on.
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TROUBLESHOOTING

IF YOU SEE	TRY
All your hypotheses are about software	Add at least one about hardware and one about the environment (heat, power, dust). The clue about the bed points at airflow.
Your first test changes two things at once	Split it into two steps. Reinstalling the app and moving the laptop to a desk are separate tests.
You cannot think of what changed	Ask for dates. Installs, updates, a new location, a new charger and a new user are the usual answers.
The plan has no stop point	Add one. An example: if temperatures stay above 95 °C at idle after cleaning, stop and get a professional repair.

Safety, Tools, and Handling Hardware

Open a desktop, name what is inside, and close it without leftovers.

OBJECTIVES

- OBJ-1** Ground yourself and set up an anti-static workspace.
- OBJ-2** Open a desktop case and photograph it before disassembly.
- OBJ-3** Identify every internal component by name.
- OBJ-4** Close the case with no loose screws and every cable seated.

MATERIALS & TOOLS

- | | |
|--|---|
| ☐ Desktop tower you are allowed to open | ☐ Flashlight |
| ☐ #2 Phillips screwdriver | ☐ Phone camera |
| ☐ Anti-static wrist strap | ☐ Cable ties (optional) |
| ☐ Magnetic or divided parts tray | ☐ A current backup of the machine's data |

PROCEDURE

A Prepare

- A.1** Confirm you have a backup made within the last day. Hardware work sometimes goes wrong.

CAUTION Never open the power supply unit itself. Its capacitors can hold a dangerous charge long after it is unplugged.

- A.2** Shut down, turn off the PSU switch, unplug the power cord, then hold the power button for five seconds to drain what is left.
- A.3** Clear a hard, well-lit table and keep carpet out of the area. Clip the wrist strap to bare metal on the case frame.

B Open and photograph

- B.1** Remove the side panel screws at the back and put them in tray section 1. Slide the panel back and lift it off. Most side panels open away from the rear I/O ports.
- B.2** Before you touch anything, photograph the whole interior, then close-ups of every cable connection.

C Identify

- C.1** Point to and name each part: motherboard, CPU cooler, RAM sticks (count them), storage drives (SATA or M.2), graphics card if present, power supply, case fans, and the front-panel header cables.
- C.2** For each cable, follow it from the PSU to where it ends and name the connector: 24-pin ATX, 8-pin CPU (EPS), PCIe power, SATA power.
- C.3** Add the internal inventory to your machine record from Lab 1.

D Close

- D.1** Compare the interior with your first photo. Any connector you touched should be fully seated, and no cable should rest against a fan.
- D.2** Refit the panel. Count the screws: the tray should be empty. Plug in, switch the PSU on, and confirm the machine boots normally.

TROUBLESHOOTING

IF YOU SEE	TRY
The side panel won't slide	Look for a second screw or a latch at the top or bottom. Some cases have a tool-less handle behind a small lip.
No power after closing	Check the PSU rocker switch first, then check that the front-panel power switch header was not pulled loose.
Fans louder than before	A cable may be touching a fan blade. Reopen the case and use a cable tie to route it away.
You have no wrist strap	Touch bare metal on the case before handling each part and keep one hand on the frame. A strap is still better.

The Processor

Read your CPU's specifications and watch it work under load.

OBJECTIVES

- OBJ-1** Find your CPU's model, cores, threads, clock speeds and cache sizes.
- OBJ-2** Explain what each specification means for real-world performance.
- OBJ-3** Watch per-core usage, clock speed and temperature under load.
- OBJ-4** Recognise the signs that the CPU is slowing itself down to stay cool (thermal throttling).

MATERIALS & TOOLS

- ☐ Any computer
- ☐ The manufacturer's spec page for your CPU model
- ☐ A CPU-heavy task such as a video export, a benchmark, or the loop in step C.1

PROCEDURE

A Identify

- A.1** Get the processor model and its core and thread counts.

WIN	PowerShell: <code>Get-CimInstance Win32_Processor Select Name,NumberOfCores,NumberOfLogicalProcessors,MaxClockSpeed</code>
MAC	<code>sysctl -n machdep.cpu.brand_string ; sysctl hw.physicalcpu hw.logicalcpu</code>
LINUX	<code>lscpu</code>

- A.2** Look up the model on the manufacturer's spec page. Write down base and boost clock, L2 and L3 cache, socket, and instruction architecture (x86-64 or ARM64).

- A.3** Confirm the OS is 64-bit.

WIN	Settings > System > About > System type
MAC	<code>uname -m</code> (arm64 or x86_64)
LINUX	<code>uname -m</code>

B Interpret

- B.1** Next to each number, write one line on what it means in practice. For example: 8 threads means eight tasks can run at the same moment. A larger L3 cache means fewer slow trips to RAM.

- B.2 Write down why a newer 3.0 GHz chip can beat an older 4.0 GHz one: it does more work per clock cycle.

c Load and observe

- C.1 Open the per-core view, then start a CPU-heavy task.

WIN	Task Manager > Performance > CPU > right-click graph > Logical processors
MAC	Activity Monitor > Window > CPU History
LINUX	htop (or: top then press 1)

- C.2 If you don't have a heavy task, run a simple loop on each core for 60 seconds, then stop it with Ctrl+C.

WIN	PowerShell: 1..\$env:NUMBER_OF_PROCESSORS % { Start-Job { while(\$true){} } } then: Get-Job Stop-Job
MAC	yes > /dev/null & (repeat per core) then: killall yes
LINUX	yes > /dev/null & (repeat per core) then: killall yes

CAUTION Do this on a machine with working fans. Stop at once if the temperature goes above 95 °C.

- C.3 Write down whether usage spreads across all cores or sits on one. Also note whether the clock speed drops as time goes on. A steady drop while the load stays the same points to thermal throttling.

TROUBLESHOOTING

IF YOU SEE	TRY
Only one core hits 100%	The program uses a single thread. More cores won't help it; a higher clock speed or a newer CPU will.
Clock speed drops after 30 seconds of load	This is thermal or power throttling. Check the cooler and dust (Lab 10), and on laptops check the power plan.
CPU at 100% while idle	A runaway process. Sort by CPU in the task manager and see Lab 18.
The OS shows half the threads you expected	Hyper-threading or SMT may be turned off in the firmware settings. Check them (Lab 9).

Memory (RAM)

Check capacity and usage, then run a full memory test.

OBJECTIVES

- OBJ-1** Find how much RAM is installed, its speed, its generation (DDR type) and which slots are filled.
- OBJ-2** Tell the difference between memory pressure and faulty memory.
- OBJ-3** Run a full memory test and read the results.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ One to eight hours for a full test pass
- ☐ A USB flash drive of 1 GB or more (for the bootable test)

PROCEDURE

A Inventory

- A.1** Find the installed capacity, speed and slot usage.

WIN	Task Manager > Performance > Memory (speed, slots used, form factor)
MAC	> About This Mac ; system_profiler SPMemoryDataType
LINUX	free -h ; sudo dmidecode --type memory grep -E 'Size Speed Type: Locator'

- A.2** Write down whether the sticks are installed in matched pairs, and how many slots are empty. Apple silicon Macs and many thin laptops have memory soldered to the board, so there are no slots to fill.

B Read pressure

- B.1** With your normal set of apps open, write down memory used, compressed memory and swap (or page file) in use.

WIN	Task Manager > Memory > In use (Compressed), Committed
MAC	Activity Monitor > Memory > Memory Pressure graph, Swap Used
LINUX	free -h ; vmstat 1 5 (watch the si/so columns)

- B.2** Decide which case you are in. Running out of memory means usage stays high, swap is active, and the machine is slow but stable. Failing memory means random crashes and errors while usage is normal.

C Test

C.1 Start the built-in or bootable memory test.

WIN	Run: mdsched.exe > Restart now and check
MAC	Apple silicon: shut down, hold power until Startup Options, then Cmd+D. Intel: hold D at startup
LINUX	Choose Memtest86+ from the GRUB menu, or boot a Memtest86+ USB

CAUTION Save your work before starting. The test reboots the machine and it cannot be used while the test runs.

C.2 For a stronger test, let MemTest86 or Memtest86+ run at least one full pass. More passes are better; four is a good target.

C.3 Record the result. Any error at all means memory is failing. To find which stick is bad, test one stick at a time.

TROUBLESHOOTING

IF YOU SEE	TRY
The test reports errors	Reseat the sticks and test again. If errors remain, test each stick on its own in the same slot to find the bad one.
Installed RAM is higher than usable RAM	Integrated graphics reserves part of the memory for itself. On a 32-bit OS, the limit is about 4 GB.
A new stick isn't detected	Check the DDR generation matches the board and the stick is fully latched on both ends. Then check the board's supported-memory list.
The system runs in single-channel mode	Move the sticks into the colour-coded paired slots listed in the board manual, usually A2 and B2.

Storage: Drives and File Systems

Check drive health and know what kind of drive you have.

OBJECTIVES

- OBJ-1** Find out whether each drive is an HDD or an SSD, and how it connects (SATA or NVMe).
- OBJ-2** Read the drive's SMART health data and interpret the key attributes.
- OBJ-3** Check how full each volume is and which file system it uses.
- OBJ-4** Know which readings mean you should back up immediately.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A backup destination in case the results are bad
- ☐ smartmontools (free) — or CrystalDiskInfo on Windows

PROCEDURE

A Identify drives

- A.1** List the physical drives with their type and connection.

WIN	PowerShell: Get-PhysicalDisk Select FriendlyName,MediaType,BusType,Size,HealthStatus
MAC	diskutil list ; system_profiler SPNVMeDataType SPSerialATADataType
LINUX	lsblk -d -o NAME,ROTA,TRAN,SIZE,MODEL (ROTA 1 = spinning)

- A.2** List the volumes, the file system on each, and free space.

WIN	Run: diskmgmt.msc
MAC	diskutil list ; df -h
LINUX	lsblk -f ; df -hT

B Read health

B.1 Read the SMART data for each drive.

WIN	CrystalDiskInfo –or– smartctl -a /dev/sda (smartmontools)
MAC	diskutil info disk0 grep SMART ; smartctl -a disk0 (brew install smartmontools)
LINUX	sudo smartctl -a /dev/sda (or /dev/nvme0)

B.2 For a hard drive, write down Reallocated Sectors, Current Pending Sectors and Uncorrectable Sectors. Any of these above zero, or rising, is a warning. For an SSD, write down Percentage Used (or Wear Leveling), Available Spare and Media Errors.

B.3 Write down the power-on hours and compare them with the drive's age. A drive that has been running constantly ages faster than one used a few hours a day.

C Decide

C.1 Mark each drive as healthy, watch, or back up now. Any warning attribute that is rising, clicking noises, or read errors in the logs means back up now.

CAUTION Do not run defragmenters, surface scans or benchmarks on a drive that shows errors. Copy your data off it first.

C.2 Add the drive details and today's SMART values to your machine record so you can compare against them later (Lab 29).

TROUBLESHOOTING

IF YOU SEE	TRY
SMART shows unavailable	Drives inside USB enclosures often block SMART. Connect the drive internally, or try smartctl -d sat.
A healthy status but the drive is very slow	Check free space. A drive that is more than 90% full slows down, SSDs especially. Also check for a SATA drive running in an old compatibility mode.
A new drive doesn't appear in Explorer or Finder	It has not been set up yet. Initialise and format it in Disk Management or Disk Utility, after confirming it is the correct, empty drive.
NVMe drive runs hot and slows down	Fit the heatsink or thermal pad supplied with the board. Controllers above 70 °C often throttle.

The Motherboard and Firmware

Tour the firmware setup safely and find the CMOS battery.

OBJECTIVES

- OBJ-1** Find your board model and firmware version.
- OBJ-2** Enter the firmware setup (BIOS or UEFI) and find the key settings without changing anything.
- OBJ-3** Find the CMOS battery and know the symptoms of it running flat.
- OBJ-4** Read a board's slots and headers from its manual.

MATERIALS & TOOLS

- ☐ Any computer (a desktop is best for the battery step)
- ☐ The motherboard or system manual (PDF from the manufacturer)
- ☐ Phone camera
- ☐ Anti-static strap (for step C.1)

PROCEDURE

A Identify board and firmware

- A.1** Find the board model, firmware version and boot mode.

WIN	Run: msinfo32 > BaseBoard Product, BIOS Version/Date, BIOS Mode
MAC	system_profiler SPHardwareDataType (Model Identifier, System Firmware Version)
LINUX	sudo dmidecode -t baseboard -t bios ; [-d /sys/firmware/efi] && echo UEFI echo Legacy

- A.2** Download the board manual. Find the layout diagram and label on it: CPU socket, RAM slots, PCIe slots, M.2 slots, SATA ports, front-panel header, fan headers and CMOS battery.

B Tour the firmware setup

B.1 Restart into the firmware setup.

WIN	Settings > System > Recovery > Advanced startup > Restart now > Troubleshoot > Advanced options > UEFI Firmware Settings
MAC	No user firmware menu. Apple silicon: shut down, hold power for Startup Options. Intel: Cmd+R for Recovery > Startup Security Utility
LINUX	systemctl reboot --firmware-setup (or press Del / F2 / F10 at the logo)

CAUTION Look only; don't change anything. Leave the setup with Exit without saving.

B.2 Find and photograph: the boot order, SATA mode (AHCI or RAID), Secure Boot, the memory profile (XMP or EXPO), the virtualisation setting, and the fan curve or hardware monitor page.

B.3 Write down the CPU and board temperatures shown in the hardware monitor. This is your baseline for Lab 10.

C CMOS battery

C.1 Power down and unplug (Lab 5 procedure). Find the coin cell on the board, usually a CR2032, and photograph it. Don't remove it.

C.2 Write down the classic symptoms of a flat CMOS battery: the clock resets at every cold start, firmware settings return to defaults, and the machine shows a "CMOS checksum" message.

TROUBLESHOOTING

IF YOU SEE	TRY
The setup key doesn't work	Fast boot skips the key prompt. Use the OS route in step B.1, or hold the key down from the moment you press power.
The OS reports Legacy boot mode on modern hardware	Changing to UEFI mode needs a drive converted from MBR to GPT. Don't switch without planning; see Lab 26.
Clock wrong after every cold start	Replace the CR2032. Then check the time and boot order in the firmware setup, because both may have reset.
Settings changed and now it won't boot	Choose Load optimized defaults, or clear CMOS with the jumper or the button shown in the manual.

Power, Cooling, and the Physical Environment

Measure temperatures, clean out dust, and check laptop battery health.

OBJECTIVES

- OBJ-1** Record idle and load temperatures and compare them.
- OBJ-2** Clean the dust out of a machine safely.
- OBJ-3** Read a laptop battery's design capacity, full charge capacity and cycle count.
- OBJ-4** Match power and heat symptoms to their likely causes.

MATERIALS & TOOLS

- ☐ Any computer; a laptop for part C
- ☐ Compressed air or an electric duster
- ☐ Soft brush
- ☐ Isopropyl alcohol (90%+) and lint-free cloth
- ☐ Temperature tool: HWiNFO (Windows), lm-sensors (Linux)
- ☐ Screwdriver matching the case or laptop base

PROCEDURE

A Temperatures

- A.1** Record the CPU temperature after 10 minutes of idle.

WIN	HWiNFO > Sensors (CPU Package)
MAC	Third-party (e.g. iStat Menus); on Intel: <code>sudo powermetrics --samplers smc</code>
LINUX	<code>sudo apt install lm-sensors ; sensors</code>

- A.2** Run the load from Lab 6 for five minutes and record the peak temperature and whether the fans get louder. Typical safe ranges: idle 30–50 °C, under load below 90 °C for most desktop CPUs.

B Clean

- B.1** Power down, unplug, and remove the laptop battery if it can be removed. Open the case or the base panel.

CAUTION Hold each fan still with a finger while you blow air at it. A spinning fan can act as a generator and damage the board.

B.2 Blow dust out through the heatsink fins in the direction opposite normal airflow, in short bursts with the can held upright. Brush the intake filters.

B.3 Reassemble the machine and repeat part A. Record the difference in temperature.

C Battery health (laptop)

C.1 Generate the battery report.

WIN	<code>powercfg /batteryreport (opens battery-report.html)</code>
MAC	<code>System Settings > Battery > Battery Health (i) ; system_profiler SPPowerDataType grep -E 'Cycle Condition Maximum'</code>
LINUX	<code>upower -i \$(upower -e grep BAT)</code>

C.2 Calculate health as full charge capacity ÷ design capacity. Below 80%, or a cycle count above the manufacturer's rating, means plan a replacement.

CAUTION A swollen battery (the case bulges, the trackpad lifts) is a fire risk. Stop using the laptop and have the battery replaced.

TROUBLESHOOTING

IF YOU SEE	TRY
Shuts off under load but not at idle	The cause is either heat or power. If temperatures are near the limit just before shutdown, clean and repaste. If they aren't, suspect the PSU (desktop) or the charger (laptop).
Fans never spin	Check the fan header in the firmware hardware monitor. A fan that shows 0 RPM with the CPU above 60 °C has failed or is unplugged.
Random restarts with no logs	A classic sign of a failing PSU. Swap in a known-good supply before you replace anything else.
Temperatures barely changed after cleaning	The thermal paste may have dried out. Repasting is the next step, but only if you are comfortable removing the cooler.

Graphics and Display

Identify the graphics hardware and set up a display properly.

OBJECTIVES

- OBJ-1** Find whether the machine has integrated or dedicated graphics, and how much video memory it has.
- OBJ-2** Match each cable to the right port for the resolution and refresh rate you need.
- OBJ-3** Set native resolution, refresh rate and scaling.
- OBJ-4** Work out whether a display fault is in the screen, the cable or the graphics.

MATERIALS & TOOLS

- ☐ Computer and an external monitor
- ☐ A spare cable of the same type (for isolation)
- ☐ HDMI and/or DisplayPort cable (and USB-C adapter if needed)

PROCEDURE

A Identify graphics

- A.1** List the graphics adapters and video memory.

WIN	Run: dxdiag > Display tab ; Task Manager > Performance > GPU
MAC	system_profiler SPDisplaysDataType
LINUX	lspci -k grep -A3 -Ei 'vga 3d'

- A.2** Write down whether each adapter is integrated (shares system RAM) or dedicated (has its own VRAM), and which driver it is using.

B Connect and configure

- B.1** Connect the monitor. Read the monitor's manual to find its native resolution and maximum refresh rate at each input, because HDMI and DisplayPort can support different rates.
- B.2** Set native resolution, maximum refresh and a comfortable scale.

WIN	Settings > System > Display > Advanced display
MAC	System Settings > Displays (Option-click Scaled for all modes)
LINUX	Settings > Displays —or— xrandr (X11)

- B.3** If you have two displays, arrange them to match their physical positions and choose which one is the main display. Drag the cursor across the edge to confirm.

c Isolate a fault

- C.1** Unplug the cable at the monitor end and write down the monitor's message (for example, No signal). That message tells you the screen itself works.
- C.2** Swap in the spare cable, then a different port on the computer, then a different input on the monitor. Change one thing each time and write down the result.
- C.3** Write a rule: artifacts that also appear in screenshots come from the graphics hardware or driver. Artifacts that don't appear in screenshots come from the cable or the screen.

TROUBLESHOOTING

IF YOU SEE	TRY
Limited to 30 Hz at 4K	The cable or port may be HDMI 1.4. Use HDMI 2.0+ or DisplayPort, with a cable rated for the bandwidth.
No signal from a desktop	The monitor may be plugged into the motherboard output while a graphics card is installed. Plug it into the graphics card.
Text is blurry	The resolution isn't native. Set native resolution and adjust scaling instead.
Flicker only at the maximum refresh rate	Try a certified cable, or drop one refresh step. If the flicker only happens with one driver version, see Lab 17.

Input, Output, and Peripherals

Use a fixed isolation sequence to diagnose a peripheral that isn't working.

OBJECTIVES

- OBJ-1** Identify USB versions and connectors from their labels and speeds.
- OBJ-2** Apply the port → cable → device → driver isolation sequence in order.
- OBJ-3** Fix a pairing or default-device problem on audio and Bluetooth devices.

MATERIALS & TOOLS

- ☐ Computer
- ☐ A USB device (mouse, flash drive)
- ☐ Headphones or speakers
- ☐ A Bluetooth device
- ☐ A helper who secretly unplugs or misconfigures one device (optional)

PROCEDURE

A Read the ports

- A.1** List the USB devices and the speed each one has negotiated.

WIN	Device Manager > View > Devices by connection ; USBTreeView (optional)
MAC	system_profiler SPUSBDataType (look for Speed)
LINUX	lsusb -t (480M = USB 2, 5000M = USB 3)

- A.2** Move a flash drive between two ports and write down whether the reported speed changes. The blue or SS-marked ports are USB 3.

B The isolation sequence

- B.1** Ask your helper to break one peripheral without telling you how, or unplug the device and change its settings yourself.
- B.2** Port: try a different port, preferably on the machine itself rather than a hub.
- B.3** Cable: swap in a known-good cable, or try the device on another computer.
- B.4** Device: try a known-good device of the same kind in the original port.

- B.5 Driver: check the device list for warnings, then remove the device and plug it in again so the driver reinstalls.

WIN	Device Manager > right-click > Uninstall device, then replug
MAC	System Information > USB ; reset Bluetooth module via Control Center
LINUX	dmesg -w (then replug) ; lsmod

- B.6 Write down the step where the fault was found, and how many steps it took.

C Audio and Bluetooth

- C.1 Change the default output device and confirm the sound moves with it.

WIN	Settings > System > Sound > Output
MAC	System Settings > Sound > Output
LINUX	Settings > Sound –or– <code>pactl list short sinks ; pactl set-default-sink <name></code>

- C.2 Remove (forget) a Bluetooth device and pair it again from scratch. Write down every step that needed the device itself to be in pairing mode.

TROUBLESHOOTING

IF YOU SEE	TRY
The device works in one port only	That port or its internal header may be faulty, or it may not supply enough power. Use a powered hub for devices that draw a lot of power.
No sound, but the device shows as connected	The default output changed. Also check the app's own output setting and the per-app volume mixer.
The Bluetooth device pairs but disconnects	Move away from 2.4 GHz sources and USB 3 ports, which cause interference, and update the Bluetooth driver or firmware.
A USB-C cable charges but carries no data	Some USB-C cables carry power only. Use one marked for data or for 10–40 Gbps.

How It All Boots: From Button to Desktop

Document each boot stage, then pin simulated failures to a stage.

OBJECTIVES

- OBJ-1** Name the boot stages in order: power, POST, firmware handoff, bootloader, kernel, drivers, desktop.
- OBJ-2** Link each stage to its visible sign.
- OBJ-3** Given a symptom, name the stage where the boot failed and the first thing to check.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ Stopwatch
- ☐ The board or system manual (for beep and LED codes)
- ☐ The boot map table you draw in step A.1

PROCEDURE

A Build the boot map

- A.1** Draw a seven-row table with the columns Stage · What happens · What you see · What failure looks like.
- A.2** Do a cold boot (as in Lab 3) and fill in the What you see column from what you actually observe.
- A.3** Look at the logs for the boot you just did.

WIN	Event Viewer > Windows Logs > System (filter: Kernel-Boot, source 12/13)
MAC	log show --last boot --predicate 'process == "kernel"' head
LINUX	journalctl -b ; systemd-analyze blame

B Diagnose simulated failures

- B.1** For each symptom below, write the failing stage and your first check: (1) Nothing at all happens, no lights. (2) Fans spin, no picture, three beeps. (3) The logo shows, then "No bootable device". (4) The OS logo spins, then the machine restarts in a loop. (5) Login works but the desktop is black.
- B.2** Check your answers against the boot map. (Model answers: power input or PSU; POST memory fault; storage, connection or boot order; OS or driver, so try safe mode; the shell or display driver.)

C Decode your board

- C.1** Find your board's beep codes or debug LED meanings in the manual and copy them into the boot map. Keep this sheet for Lab 26.

TROUBLESHOOTING

IF YOU SEE	TRY
There are no beeps at all	Many boards have no speaker. Look for debug LEDs labelled CPU, DRAM, VGA and BOOT, or a two-digit POST code display.
Stuck on the logo for a long time	The firmware may be waiting on a slow or failing drive. Disconnect the extra drives and try again.
The boot loop is too fast to read	Turn off automatic restart on system failure (Windows: Startup Settings) so the error stays on screen.
Linux stops at a prompt such as initramfs or grub rescue	You are in the bootloader or early kernel stage. Write down the exact message, which usually names the missing disk or file.

What an Operating System Does

Tour the tools the OS uses to manage your hardware.

OBJECTIVES

- OBJ-1** Find the OS tool that manages each of: processes, memory, storage, devices, services and users.
- OBJ-2** Check your account type and what it is allowed to do.
- OBJ-3** Decide whether a sample symptom points to the OS or to the hardware.

MATERIALS & TOOLS

- ☐ Any computer with an administrator account
- ☐ This lab's tool table (you fill it in)

PROCEDURE

A Tour the management tools

- A.1** Open each tool and write one sentence on what it controls.

WIN	taskmgr · devmgmt.msc · diskmgmt.msc · services.msc · eventvwr.msc · lusrmgr.msc (Pro)
MAC	Activity Monitor · System Information · Disk Utility · Console · System Settings > Users & Groups
LINUX	top/htop · lsblk · systemctl · journalctl · id / getent group

- A.2** In the services or daemons list, find three services and write what each one does. Note which start automatically.

B Accounts and permissions

- B.1** Check whether your account is an administrator.

WIN	PowerShell: whoami /groups findstr Administrators
MAC	id -Gn grep -w admin
LINUX	id ; groups (look for sudo or wheel)

- B.2** Try to open a system file, such as the hosts file, in a plain editor and save a change. Write down what the OS says, then close without saving. This is permission enforcement at work.

C OS or hardware?

- C.1** Sort these symptoms and give a reason for each: (1) Errors only when logged in as one user. (2) Crashes even in the firmware setup. (3) Slow after an OS update. (4) Clicking noise from inside the case. (5) One app freezes; others keep working.

TROUBLESHOOTING

IF YOU SEE	TRY
A tool is missing	lsuirmgr.msc isn't available on Windows Home; use Settings > Accounts instead. The Linux tools depend on the distribution, so try the GUI Settings app.
Access denied, even as an administrator	Windows and macOS run programs with limited rights by default. Right-click > Run as administrator, or use sudo, but only for the task that needs it.
The Console or journal is overwhelming	Filter by time. Reproduce the problem, then look only at the last two minutes of entries.

Files, Folders, and Data Organization

Map your user folders and find where a program stores its data.

OBJECTIVES

- OBJ-1** Draw the folder tree of your user account.
- OBJ-2** Show file extensions and explain why hiding them is risky.
- OBJ-3** Find where one program keeps its settings and data.
- OBJ-4** Know which folders you should never delete from.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ Paper for a tree diagram
- ☐ A program you use daily (browser, note app)

PROCEDURE

A Show what is hidden

- A.1** Turn on file extensions and hidden files.

WIN	File Explorer > View > Show > File name extensions, Hidden items
MAC	Finder > Settings > Advanced > Show all filename extensions ; Cmd+Shift+. for hidden
LINUX	Files > ≡ > Show Hidden Files (Ctrl+H)

- A.2** Find a file whose real extension surprises you, and write down why an attachment named invoice.pdf.exe is dangerous when extensions are hidden.

B Map your user folder

- B.1** Print two levels of your home folder.

WIN	<code>cd %USERPROFILE% ; tree /A more</code>
MAC	<code>cd ~ ; find . -maxdepth 2 -type d head -60</code>
LINUX	<code>cd ~ ; tree -L 2 (or find . -maxdepth 2 -type d)</code>

- B.2** Draw the tree. Mark the folders that hold your files (Documents, Desktop, Pictures), the folders that hold program data, and the folders you should leave alone.

C Find a program's data

C.1 Go to the application data location and find your chosen program's folder.

WIN	%APPDATA% and %LOCALAPPDATA%
MAC	~/Library/Application Support and ~/Library/Preferences
LINUX	~/.config and ~/.local/share

C.2 Write the full path. Note the folder's size, and whether it contains a cache you could safely clear.

C.3 Search for a file by name and then by content, and time both searches. Write down which search tool you would trust for a lost-file emergency.

TROUBLESHOOTING

IF YOU SEE	TRY
A file opens in the wrong program	Change the default: right-click › Open with › Choose another app (Windows), Get Info › Open with › Change All (macOS), or Properties › Open With (Linux).
A shortcut or alias opens nothing	The target was moved or deleted. A shortcut only points to a file; deleting the shortcut doesn't delete the file.
You can't find Library on macOS	It is hidden. Use Finder › Go › Go to Folder and type ~/Library.
The drive is full and you don't know why	Use a space visualiser (WinDirStat, GrandPerspective, Baobab) before deleting anything.

Installing, Updating, and Removing Software

Audit installed software and startup items, and remove one thing cleanly.

OBJECTIVES

- OBJ-1** List everything installed and everything that starts automatically at login.
- OBJ-2** Tell trusted install sources from risky ones.
- OBJ-3** Uninstall one program and check for leftovers.
- OBJ-4** Measure the effect of cleaning up startup items.

MATERIALS & TOOLS

- ☐ Any computer with administrator access
- ☐ A program you no longer need (or install a small free one first)
- ☐ Stopwatch

PROCEDURE

A Audit

- A.1** List the installed software.

WIN	Settings > Apps > Installed apps –or– winget list
MAC	/Applications ; brew list (if Homebrew) ; App Store > Account
LINUX	apt list --installed wc -l ; flatpak list ; snap list

- A.2** List what starts automatically.

WIN	Task Manager > Startup apps (note Startup impact)
MAC	System Settings > General > Login Items & Extensions
LINUX	ls ~/.config/autostart ; systemctl list-unit-files --state=enabled

- A.3** Mark each startup item as keep, disable, or unknown. Look up each unknown item before deciding.

B Clean up

- B.1** Time a cold boot to a usable desktop. Disable two startup items marked disable, reboot, and time it again.

B.2 Uninstall the program you don't need using the official route.

WIN	Settings > Apps > ... > Uninstall –or– winget uninstall <id>
MAC	Drag to Trash; check the vendor's uninstaller if one was supplied
LINUX	sudo apt remove <pkg> ; flatpak uninstall <app>

B.3 Check for leftovers in the application data paths from Lab 15 and remove only folders clearly named after that program.

C Update

C.1 Run the system update and write down what was offered.

WIN	Settings > Windows Update ; winget upgrade --all
MAC	System Settings > General > Software Update
LINUX	sudo apt update && sudo apt upgrade

TROUBLESHOOTING

IF YOU SEE	TRY
An installer offers extra software	Choose Custom or Advanced install and untick the extras. If there is no way to opt out, find a different source for the program.
You disabled an item and something stopped working	Turn it back on. Change one item at a time so you always know which one caused the problem.
The uninstaller fails	Run the repair option first, then uninstall. As a last resort, use the vendor's removal tool.
Slower after updating	Updates often index, compile or optimise in the background for about a day. Measure again after 24 hours before you blame the update.

Drivers and Device Management

Inspect devices, find flagged ones, and update or roll back a driver.

OBJECTIVES

- OBJ-1** Read the device list and its warning symbols.
- OBJ-2** Find a device's driver version and who supplied it.
- OBJ-3** Update or roll back a driver safely, and confirm the result.
- OBJ-4** Tell a driver fault from a hardware failure.

MATERIALS & TOOLS

- ☐ Any computer with administrator access
- ☐ A restore point or backup (Windows)
- ☐ The manufacturer's support page for your graphics, network or audio hardware

PROCEDURE

A Inspect

- A.1** Open the device list and look for warnings.

WIN	devmgmt.msc (yellow ! = problem, ↓ = disabled)
MAC	System Information > Hardware and Software > Extensions ; systemextensionsctl list
LINUX	lspci -k (Kernel driver in use) ; lsusb ; dmesg --level=err,warn

- A.2** For your graphics and network adapters, write down the driver name, version and date or source.

B Update or roll back

- B.1** Make a restore point first.

WIN	Search: Create a restore point > Create
MAC	Time Machine snapshot: tmutil localsnapshot
LINUX	Timeshift snapshot, or keep the previous kernel in GRUB

- B.2 Update one driver from the manufacturer's site, or roll it back if it was recently updated and is causing trouble.

WIN	Device > Properties > Driver > Update driver / Roll Back Driver
MAC	Drivers come with macOS updates; remove any third-party system extension if flagged
LINUX	Update via package manager; to roll back a kernel driver, boot the previous kernel from GRUB > Advanced options

CAUTION Change one driver at a time. Reboot and test before you touch the next one.

- B.3 Reboot, test the device, and log the version change and the result.

c Driver or hardware?

- C.1 Write a two-line test for telling them apart. First, does the fault also happen in a live USB OS or in safe mode with generic drivers? Second, does it follow the device to another machine? If the answer to both is no, the driver is the likely cause.

TROUBLESHOOTING

IF YOU SEE	TRY
There is no Roll Back option	This is the first driver installed. Uninstall it, tick Delete the driver software, and let Windows install its generic driver.
Windows Update keeps reinstalling the bad driver	Use the Show or hide updates troubleshooter, or change the Device installation settings.
Wi-Fi is gone after a Linux kernel update	The firmware or module may be missing for the new kernel. Boot the previous kernel and check <code>dmesg grep -i firmware</code> .
An Unknown device is listed	Open Properties > Details > Hardware IDs and search for the VEN and DEV values to identify it.

Processes, Memory, and Performance

Use the task manager to find the cause of a slowdown.

OBJECTIVES

- OBJ-1** Read CPU, memory, disk and network usage for each process.
- OBJ-2** Find which of the four resources is the bottleneck.
- OBJ-3** End a runaway process safely.
- OBJ-4** Tell a temporary slowdown (updates, indexing) from a persistent one.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A reproducible slowdown, or create one: 30 browser tabs plus a large file copy

PROCEDURE

A Baseline

- A.1** At idle, record total CPU, memory, disk and network usage.

WIN	Task Manager > Performance ; resmon for detail
MAC	Activity Monitor (CPU, Memory, Disk, Network tabs)
LINUX	htop ; iotop (sudo) ; nload or nethogs

B Create and locate the bottleneck

- B.1** Reproduce the slowdown. Find which resource is at or near its limit. Disk at 100% active time counts, even when the amount transferred is small.
- B.2** Sort the processes by that resource and write down the top one. Also note its parent program, which you can see in the Details view or with pstree.
- B.3** Check for memory pressure specifically.

WIN	Task Manager > Memory: Committed near limit, Hard faults/sec in resmon
MAC	Activity Monitor > Memory Pressure yellow/red, Swap Used rising
LINUX	vmstat 1 : si/so columns non-zero and rising

C Act

C.1 Close the program normally first. If it is stuck, end the process.

WIN	Task Manager › right-click › End task
MAC	Activity Monitor › ⌘ › Quit, then Force Quit
LINUX	kill <PID>, then kill -9 <PID> only if needed

CAUTION Never end processes you don't recognise just to free memory. Look them up first; some keep the system running.

C.2 Write the diagnosis in one sentence: resource · process · cause · fix.

TROUBLESHOOTING

IF YOU SEE	TRY
Disk at 100% right after startup	Updates, indexing and antivirus scans do this. Wait 15 minutes. If it continues, check the drive's health (Lab 8).
High CPU from a system process	On Windows, look up the service hosted inside svchost in the Services column. On macOS, kernel_task is usually heat management, so check temperatures.
Memory full, but no single app is large	Many small processes add up, especially browser tabs. Close tabs or add RAM (Lab 7).
The network is saturated	Look for cloud sync, updates or backups running. Pause them and test again.

The Command Line

Work through a guided set of navigation and diagnostic commands.

OBJECTIVES

- OBJ-1** Navigate folders and manage files from a terminal.
- OBJ-2** Run the built-in diagnostic commands and read their output.
- OBJ-3** Read a command you were given and understand it before running it.
- OBJ-4** Use elevated permissions only when needed.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A scratch folder you can delete afterwards
- ☐ Terminal: PowerShell (Windows), Terminal (macOS), any shell (Linux)

PROCEDURE

A Navigate

- A.1** Show where you are, list what is there, and move around.

WIN	<code>Get-Location ; Get-ChildItem ; cd Documents ; cd ..</code>
MAC	<code>pwd ; ls -la ; cd Documents ; cd ..</code>
LINUX	<code>pwd ; ls -la ; cd Documents ; cd ..</code>

- A.2** Create a scratch folder, then create, copy, rename and delete a file inside it.

WIN	<code>mkdir lab19 ; cd lab19 ; ni a.txt ; copy a.txt b.txt ; ren b.txt c.txt ; del c.txt</code>
MAC	<code>mkdir lab19 && cd lab19 ; touch a.txt ; cp a.txt b.txt ; mv b.txt c.txt ; rm c.txt</code>
LINUX	<code>mkdir lab19 && cd lab19 ; touch a.txt ; cp a.txt b.txt ; mv b.txt c.txt ; rm c.txt</code>

CAUTION Deleting from the command line bypasses the Recycle Bin or Trash. Check the file name before pressing Enter.

B Diagnose

B.1 Network check.

WIN	ipconfig ; ping -n 4 1.1.1.1 ; nslookup example.com
MAC	ifconfig en0 ; ping -c 4 1.1.1.1 ; nslookup example.com
LINUX	ip addr ; ping -c 4 1.1.1.1 ; nslookup example.com

B.2 Disk and system check.

WIN	Get-Volume ; systeminfo more
MAC	df -h ; uptime ; sw_vers
LINUX	df -h ; uptime ; uname -a

B.3 Integrity check (elevated).

WIN	Admin terminal: sfc /scannow ; DISM /Online /Cleanup-Image /ScanHealth
MAC	diskutil verifyVolume /
LINUX	sudo fsck -n /dev/sdXN (unmounted volume only)

C Read before you run

C.1 Take this command from a help article: `Remove-Item -Recurse -Force C:\Temp\*` (or `rm -rf /tmp/old/*`). Write what each part does and what could go wrong if the path were mistyped. Don't run it.

C.2 Use the built-in help to check one flag.

WIN	Get-Help Remove-Item -Examples
MAC	man rm
LINUX	man rm --or- rm --help

TROUBLESHOOTING

IF YOU SEE	TRY
Command not recognised	Check the spelling and which shell you are in. PowerShell and cmd use different commands. On Linux, the tool may not be installed.
Permission denied	The action needs elevation. Rerun that single command with an admin terminal or sudo.
A path with spaces fails	Wrap it in quotes: <code>cd "My Documents"</code> .
The output scrolls past too fast	Pipe it to more or less, or save it to a file: <code>command > out.txt</code> .

Backups, Recovery, and Protecting Your Data

Set up a working backup, then prove it by restoring a file.

OBJECTIVES

- OBJ-1** Tell backup, sync and archive apart.
- OBJ-2** Set up an automatic backup that follows the 3-2-1 rule.
- OBJ-3** Delete a test file and restore it from the backup.
- OBJ-4** Find the OS recovery tools before you need them.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A cloud account (optional, for the off-site copy)
- ☐ External drive at least as large as your data
- ☐ A test file with unique contents

PROCEDURE

A Plan

- A.1** List your data and how much loss you could tolerate for each item: an hour, a day or a week. This sets how often the backup runs.
- A.2** Write your 3-2-1 plan: three copies, on two kinds of media, with one copy off-site. Name the actual device or service for each copy.

B Set up

- B.1** Configure the built-in backup to the external drive.

WIN	Control Panel > File History > Turn on –or– Windows Backup app
MAC	System Settings > General > Time Machine > Add Backup Disk
LINUX	Déjà Dup (Backups) > Storage location –or– <code>rsync -a --delete ~/ /media/backup/home/</code>

- B.2** Run the first backup now and write down how long it took and how large it is.

C Prove it

C.1 Create lab20-test.txt containing today's date and time. Run a backup. Delete the file and empty the Recycle Bin or Trash.

C.2 Restore the file.

WIN	File Explorer > folder > Home > History > Restore
MAC	Open folder > enter Time Machine > select > Restore
LINUX	Déjà Dup > Restore –or– copy back from the rsync target

C.3 Open the restored file and confirm its contents match. Write down: backup verified, with the date.

C.4 Find the recovery environment and write down how to reach it.

WIN	Settings > System > Recovery > Advanced startup
MAC	Apple silicon: hold power > Options ; Intel: Cmd+R
LINUX	GRUB > Advanced options > recovery mode, or a live USB

TROUBLESHOOTING

IF YOU SEE	TRY
Backup drive not found	The drive may have a different letter, or it may be asleep. Use a fixed drive letter or label, and wake the drive before the backup is scheduled to run.
Sync deleted my files everywhere	Sync copies deletions to every device. It isn't a backup. Turn on version history, or add a real backup.
The backup is too slow	The first backup is always the slowest; later ones copy only changes. Exclude caches and downloads.
Restore fails with permission errors	Restore to a new folder first, then copy the files into place.

How Networks and the Internet Work

Map your home network and trace the path to a website.

OBJECTIVES

- OBJ-1** Find your device's local IP address, default gateway and DNS server.
- OBJ-2** Find your public IP and explain how it differs from your local one.
- OBJ-3** Draw your home network layout.
- OBJ-4** Trace the hops from your device to a website.

MATERIALS & TOOLS

- ☐ Any computer on a home network
- ☐ Paper for a network diagram
- ☐ Access to your router's label (for the model and default gateway)

PROCEDURE

A Your addresses

- A.1** Get your local IP address, gateway and DNS servers.

WIN	<code>ipconfig /all</code>
MAC	<code>ipconfig getifaddr en0 ; route -n get default ; scutil --dns grep nameserver</code>
LINUX	<code>ip -4 addr ; ip route ; resolvectl status</code>

- A.2** Find your public IP by visiting a what-is-my-IP site, or from a terminal.

```
curl ifconfig.me
```

- A.3** Write down why the two differ. Your router uses NAT to share one public address with every device behind it.

B Draw the network

- B.1** Draw it: internet → modem (or ONT) → router → switch or Wi-Fi → your devices. Label the gateway IP on the router and your device's IP.

- B.2** Find the other devices on the network. The easiest place is the router's connected devices page (Lab 22), or look at the table of nearby devices your machine has talked to.

WIN	<code>arp -a</code>
MAC	<code>arp -a</code>
LINUX	<code>ip neigh</code>

c Trace

- C.1** Trace the route to a website and count the hops.

WIN	<code>tracert example.com</code>
MAC	<code>tracertool example.com</code>
LINUX	<code>tracepath example.com</code> or <code>traceroute example.com</code>

- C.2** Label hop 1 (your router), the first hops at your provider, and the destination. Write down which hop has the biggest jump in delay.

- C.3** Resolve a name by hand and write the IP it returns.

<code>nslookup example.com</code>

TROUBLESHOOTING

IF YOU SEE	TRY
IP address starts with 169.254	The device didn't get an address from the router's DHCP. Restart the router and check the cable or Wi-Fi connection (Lab 23).
Traceroute shows * * * rows	Many routers don't reply to traceroute. If later hops respond, the path is fine.
Two routers in a row	This is double NAT: the provider's gateway is still routing. Consider bridge mode on one of them.
The public IP changes	Most home connections get a dynamic IP. This is normal.

Wi-Fi, Routers, and Home Networking

Log into your router, review its key settings, and work through a Wi-Fi fault.

OBJECTIVES

- OBJ-1** Log into the router's admin page.
- OBJ-2** Review security, band and channel settings, and firmware.
- OBJ-3** Measure how Wi-Fi signal and speed change from room to room.
- OBJ-4** Run a Wi-Fi troubleshooting sequence in a fixed order.

MATERIALS & TOOLS

- ☐ Home router you administer
- ☐ Router admin password (often on the label)
- ☐ Laptop or phone
- ☐ A speed test site

PROCEDURE

A Get in

- A.1** Type the gateway IP from Lab 21 into a browser (often 192.168.0.1 or 192.168.1.1) and log in. Write down the router model and firmware version.

CAUTION If the admin password is still the factory default, change it today and store it in your password manager.

B Review settings

- B.1** Check the security mode: WPA2-AES at minimum, WPA3 if every device supports it. WEP or open networks must be fixed.
- B.2** Note the SSID for each band (2.4 GHz and 5 or 6 GHz) and the channel. Turn off WPS unless you actively use it.
- B.3** Check for a firmware update. Check whether the guest network is enabled, and whether smart devices could be moved to it.

C Measure

C.1 In three rooms, record signal strength and a speed test.

WIN	netsh wlan show interfaces (Signal, Band, Channel)
MAC	Option-click Wi-Fi icon (RSSI, Channel, Tx Rate)
LINUX	nmcli -f IN-USE,SSID,CHAN,SIGNAL,RATE dev wifi

C.2 Write down which band held up best at distance, and which was fastest close to the router.

D Troubleshoot

D.1 Run this sequence and stop at the first failure. (1) Is it one device or all devices? (2) Forget the network and rejoin it on the failing device. (3) Is a wired device online? If not, the problem is the router or the internet connection. (4) Restart the modem, wait for it to come up, then restart the router. (5) Check the provider's status page.

TROUBLESHOOTING

IF YOU SEE	TRY
Can't reach the admin page	You may be on the guest network or a mesh satellite. Connect to the main SSID and use the gateway IP from ipconfig or ip route.
Fast next to the router, slow two rooms away	5 GHz loses signal through walls. Move the router higher and more central, or add a mesh node.
Frequent drops in a crowded building	Check the channel on 2.4 GHz and pick 1, 6 or 11, whichever is least used.
An old device won't connect after switching to WPA3	Use WPA2/WPA3 transitional mode, or put the old device on its own network.

Diagnosing Connection Problems

Follow the layered connection-diagnosis flowchart to the point of failure.

OBJECTIVES

- OBJ-1** Turn "the internet is down" into a specific failing layer.
- OBJ-2** Use loopback, gateway, external IP and DNS tests in order.
- OBJ-3** Tell your equipment's fault from your provider's.
- OBJ-4** Collect the facts you'll need before calling support.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ Ethernet cable (for wired checks)
- ☐ A way to simulate a fault: disable Wi-Fi, set a wrong DNS server, or unplug the router's WAN cable

PROCEDURE

A The flowchart

- A.1** Test 1, your own network stack. If this fails, reset the adapter or the network stack.

```
ping 127.0.0.1
```

- A.2** Test 2, the connection to the router. If this fails, check Wi-Fi association, the cable and the DHCP address (Lab 21).

```
ping <gateway IP>
```

- A.3** Test 3, internet by IP address. If this fails, the problem is the modem, the WAN link or the provider.

WIN	ping -n 4 1.1.1.1
MAC	ping -c 4 1.1.1.1
LINUX	ping -c 4 1.1.1.1

- A.4** Test 4, name lookup (DNS). If Test 3 passed and this fails, it is DNS.

```
nslookup example.com
```

- A.5** Test 5, the browser itself. If everything above passed, suspect the proxy, VPN, extensions or the website.

B Fix DNS

B.1 Flush the DNS cache and test again.

WIN	<code>ipconfig /flushdns</code>
MAC	<code>sudo dscacheutil -flushcache ; sudo killall -HUP mDNSResponder</code>
LINUX	<code>resolvectl flush-caches</code>

B.2 Test against a specific public DNS server: `nslookup example.com 1.1.1.1`. If it works there but not through your normal server, change the device or router DNS setting.

C Slow, not down

C.1 Run a speed test wired, then on Wi-Fi next to the router. If the wired result is good and Wi-Fi is bad, the problem is Wi-Fi. If both are bad, it is the connection or the provider.

C.2 Measure packet loss over a minute.

WIN	<code>ping -n 60 1.1.1.1</code>
MAC	<code>ping -c 60 1.1.1.1</code>
LINUX	<code>ping -c 60 1.1.1.1</code>

C.3 Write a support summary: the time the problem started, results of tests 1–5, wired speed, packet loss, and modem lights.

TROUBLESHOOTING

IF YOU SEE	TRY
Test 2 fails on Wi-Fi only	Forget the network and rejoin it. Check for MAC filtering or an old saved password.
All tests pass but websites won't load	Check the system proxy and any VPN, and try another browser. A captive portal (hotel Wi-Fi) also causes this.
Wired connection stuck at 100 Mbps	A damaged cable or one with only four wires in use. Try a Cat5e or Cat6 cable and a different switch port.
The modem's online light is off	The problem is the provider's line. Call with your summary ready.

Security, Malware, and Staying Safe

Run a scan, review the protection settings, and audit your password habits.

OBJECTIVES

- OBJ-1** Run a full malware scan and read the results.
- OBJ-2** Confirm that firewall, updates and disk encryption are on.
- OBJ-3** Audit your password reuse and turn on multi-factor authentication (MFA) on key accounts.
- OBJ-4** Write a plan for what to do if a machine is compromised.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ Your email account and phone for MFA
- ☐ A password manager (built-in or third-party)

PROCEDURE

A Scan

- A.1** Run a full scan.

WIN	Windows Security > Virus & threat protection > Scan options > Full scan
MAC	XProtect runs automatically; optional: Malwarebytes free scan
LINUX	<code>sudo apt install clamav ; freshclam ; clamscan -r --infected ~</code>

- A.2** Write down the items found and what action was taken. Look up any item before you delete it, because false positives happen.

B Protection settings

- B.1** Check that the firewall is on.

WIN	Get-NetFirewallProfile Select Name,Enabled
MAC	System Settings > Network > Firewall
LINUX	<code>sudo ufw status (or firewall-cmd --state)</code>

B.2 Check that disk encryption is on.

WIN	Settings > Privacy & security > Device encryption (or <code>manage-bde -status</code>)
MAC	System Settings > Privacy & Security > FileVault
LINUX	<code>lsblk -f</code> (look for <code>crypto_LUKS</code>)

B.3 Confirm automatic updates are on for both the OS and the browser.

c Password audit

C.1 Import or list your accounts in the password manager. Run its reused or weak password report.

C.2 Turn on MFA for email first, then banking, then the password manager itself. Use an authenticator app or a security key rather than SMS where you can.

C.3 Write your compromise plan: disconnect from the network, change the email password from a clean device, scan, and wipe and reinstall if you are unsure (Lab 20 backups make this possible).

TROUBLESHOOTING

IF YOU SEE	TRY
Scan finds a tool you installed	Security tools and key generators are often flagged. If you don't trust where it came from, remove it.
Browser pop-ups say you're infected	These are usually scam web pages, not an infection. Close the tab and don't call the number shown.
Browser homepage keeps changing	A malicious extension. Remove the unknown extensions and reset the browser settings.
Locked out after turning on MFA	Use the recovery codes you saved. Store them offline when you set up MFA.

The Diagnostic Framework in Full

Take a fault with several symptoms all the way to its root cause, and document how you got there.

OBJECTIVES

- OBJ-1** Narrow a fault step by step: system → subsystem → component → cause.
- OBJ-2** Settle whether the fault is in hardware or software with one decisive test.
- OBJ-3** Use swaps with known-good parts and the logs as evidence.
- OBJ-4** Build your personal troubleshooting checklist.

MATERIALS & TOOLS

- ☐ The fault scenario below
- ☐ A live USB of any Linux distribution (for the hardware-or-software test)
- ☐ Your boot map (Lab 13) and troubleshooting log (Lab 4)

PROCEDURE

A The fault

- A.1** Scenario: a desktop freezes a few times a day, sometimes shows a blue screen that mentions memory management, and one file recently wouldn't open. The owner upgraded the graphics driver and added a second RAM stick three weeks ago.

B Front-load the answer

- B.1** Answer three questions from the scenario. What changed? When did it start? Can it be reproduced? List the two changes and what each could cause.

- B.2** Collect the evidence you would gather: the crash logs and the file error.

WIN	<code>perfmon /rel</code> (Reliability Monitor) ; Event Viewer > System > BugCheck
MAC	Console > Crash Reports ; <code>log show --last 1d --predicate 'eventMessage contains "panic"'</code>
LINUX	<code>journalctl -b -1 -p err ; ls /var/crash</code>

C Narrow

- C.1 Hardware or software? Boot the live USB and run it under load for an hour. If it freezes there too, the fault is hardware, because the installed OS and drivers aren't running.
- C.2 Rank the causes. The memory-management error, the corrupted file and the new RAM all point to memory. The graphics driver fits the freezes but not the corrupted file.
- C.3 Test one change: remove the new stick and run for two days. Then test the new stick alone with Memtest86+ (Lab 7).
- C.4 If that doesn't settle it, substitute: put the stick in a known-good board, or a known-good stick in this board. See which one the fault follows.

D Document and generalise

- D.1 Write the fault up as: symptom → evidence → hypotheses → tests → root cause → fix → how you confirmed it.
- D.2 Turn your process into a ten-line checklist you could use on any fault. Keep it at the front of this guide.

TROUBLESHOOTING

IF YOU SEE	TRY
Two causes seem equally likely	Test the cheaper, quicker, reversible one first.
The fault won't reproduce on demand	Increase exposure: run it longer and under heavier load, and log the time of every occurrence.
The logs mention a driver file (.sys)	Look up the file name to find which driver it belongs to. It is often a strong lead, but faulty memory can make any driver crash.
The fix worked, but you don't know why	Undo the fix and confirm the fault returns. Only then is the cause proven.

When It Will Not Turn On or Boot

Work out which boot stage failed in a set of staged no-boot scenarios.

OBJECTIVES

- OBJ-1** Test the no-power path in order: wall, cable, supply, board.
- OBJ-2** Decode POST failures from beeps and LEDs.
- OBJ-3** Fix a missing boot device and a boot loop.
- OBJ-4** Get data off a machine that won't boot.

MATERIALS & TOOLS

- ☐ Your boot map (Lab 13)
- ☐ A bootable USB of your OS or a Linux live USB
- ☐ A test desktop you can open (optional)
- ☐ External drive for data rescue

PROCEDURE

A Staged scenarios

- A.1** For each, name the stage and your first three checks in order. (1) No lights, no fans. (2) Fans spin, no display, the DRAM LED is lit. (3) Reboot and select proper boot device. (4) The OS logo appears, then Automatic Repair runs again and again. (5) Boots to a black screen with a cursor.
- A.2** If you have a test desktop, stage scenarios 1–3 on it: switch off the PSU, pull one RAM stick out partway, and unplug the SATA data cable. Then diagnose them cold.

B Recovery tools

- B.1** Boot into safe mode or the recovery environment and explore it.

WIN	Hold Shift + Restart > Troubleshoot > Startup Settings > Safe Mode
MAC	Apple silicon: hold power > hold Shift on the volume for Safe Mode
LINUX	GRUB > Advanced options > recovery mode

- B.2** Find the startup repair and system restore options, but don't run them. Write down what each one changes.

C Rescue data

C.1 Boot the live USB, mount the internal drive and copy your user folder to the external drive.

WIN	Linux live USB can read NTFS; BitLocker drives need the recovery key
MAC	Recovery > Disk Utility (mount) > Terminal: <code>cp -R</code> , or Target Disk Mode / Share Disk
LINUX	Live USB > Files > internal drive > copy

CAUTION If the drive clicks or reports read errors, copy the most important folders first and stop as soon as you have them. Every hour it runs could be its last.

C.2 Write the repair-or-reinstall decision in one sentence, only after your data is safe.

TROUBLESHOOTING

IF YOU SEE	TRY
Scenario 1: no lights	Check the wall socket with a lamp, the PSU switch, and the power cable. Then short the power switch pins directly to test the front-panel button.
Scenario 3: no boot device	Check the drive appears in the firmware setup, then check the boot order, then whether the drive was set up in UEFI or legacy mode.
Scenario 4: repair loop	Try safe mode. Then uninstall the latest update or driver. Check the drive's SMART data, because this is often a dying drive in disguise.
BitLocker asks for a recovery key	Find it at the Microsoft account's recovery key page (account.microsoft.com). Don't reset the TPM until you have it.

When It Is Slow, Freezing, or Crashing

Diagnose a slowdown and an intermittent crash, and decide whether each is hardware or software.

OBJECTIVES

- OBJ-1** Measure a slowdown instead of just describing it.
- OBJ-2** Tell apart freezing, crashing and spontaneous restarting.
- OBJ-3** Confirm or rule out overheating.
- OBJ-4** Read a blue screen or kernel panic for evidence.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A notebook for timing and a log of each event
- ☐ Tools from Labs 6, 7, 10 and 18

PROCEDURE

A Measure the slowdown

- A.1** Pick one slow action (opening an app, booting) and time it three times. Record the average.
- A.2** Run the Lab 18 routine: find the bottleneck resource, then the top process. Record the temperature as well (Lab 10).
- A.3** Make one targeted change, time the action again, and log the result.

B Investigate an intermittent crash

- B.1** Classify each event: a freeze (the display is stuck), a crash to an error screen, or a sudden restart with no error. Write down the pattern each suggests: freezes point to drivers or storage, error screens to drivers or memory, and sudden restarts to power or heat.

- B.2** Pull the crash history.

WIN	perfmon /rel ; Event Viewer > System > Kernel-Power 41, BugCheck 1001
MAC	Console > Crash Reports / System Reports (panic-full-*.panic)
LINUX	journalctl --list-boots ; journalctl -b -1 -e

- B.3** Put the events in a table: time · what was running · temperature · error text. Look for what they have in common.

C Decide the intervention

- C.1** Choose the level of intervention and justify it: clean up, then repair (sfc or disk check), then reinstall, then replace hardware. Go to the next level only when the evidence rules out the current one.

TROUBLESHOOTING

IF YOU SEE	TRY
Crashes only when gaming or rendering	Load-related, so suspect heat, the PSU or the graphics driver. Monitor temperatures and power during the load.
Kernel-Power 41 with no BugCheck	The power was cut suddenly. Suspect the PSU, the wall power, or an overheating shutdown.
Error text differs every time	Varied, random errors point strongly to faulty memory. Run Memtest86+ overnight.
Slow only after the laptop has been on for an hour	Heat build-up. Check temperatures over time and clean the fans.

When a Program or Device Misbehaves

Work through program and peripheral faults with the right isolation sequence for each.

OBJECTIVES

- OBJ-1** Tell a fault in one program from a system-wide fault.
- OBJ-2** Apply program fixes from least to most drastic: restart, update, reset, reinstall.
- OBJ-3** Work printing, audio and display problems through to a fix.
- OBJ-4** Tell whether the problem is the file or the program.

MATERIALS & TOOLS

- ☐ Any computer
- ☐ A program you can safely reset
- ☐ A printer, speakers or headphones, and a second display if available

PROCEDURE

A Program ladder

- A.1** Restart the program, then restart the computer.
- A.2** Update the program. Then reset its settings by moving the settings folder aside rather than deleting it.

WIN	Settings > Apps > ... > Advanced options > Repair / Reset ; or rename %APPDATA%\<app>
MAC	Quit app ; move ~/Library/Preferences/<bundle>.plist to Desktop
LINUX	Quit app ; mv ~/.config/<app> ~/.config/<app>.bak
- A.3** Reinstall the program. Only after that, clear all its data, and only after checking the data is backed up.
- A.4** Test with a different file and a new user account. If it works for the new user, the fault is in the old profile. If only one file fails, the file is the problem.

B Printing

B.1 Clear the queue and restart the print service.

WIN	Admin: net stop spooler ; del %systemroot%\System32\spool\PRINTERS* ; net start spooler
MAC	Printers & Scanners › right-click list › Reset printing system
LINUX	sudo systemctl restart cups ; open http://localhost:631

B.2 Print a test page from the printer's own menu. If that works, the fault is in the computer, its driver or the queue. If it doesn't, the fault is in the printer.

C Rolling back after an update

C.1 Find the most recent update and roll it back, or reinstall the previous version from the vendor's archive.

C.2 Record the version that works and pause automatic updates for that program until a fix is released.

TROUBLESHOOTING

IF YOU SEE	TRY
The program won't close	End it with the task manager (Lab 18). If it keeps restarting, a service or login item is relaunching it.
It works as administrator only	The settings or data folder has the wrong permissions. Reset its ownership rather than always running as administrator.
A setting changed and nobody knows which one	Compare with a fresh user account, or restore the settings file from a backup.
The printer shows offline but is on	Its IP may have changed. Reserve an address for it in the router, or add it again by hostname.

Preventive Maintenance and Keeping a Machine Healthy

Build and run a complete maintenance checklist.

OBJECTIVES

- OBJ-1** Write a maintenance schedule: monthly, quarterly and yearly.
- OBJ-2** Run a full maintenance pass and record the results.
- OBJ-3** Compare drive health against your earlier baseline.
- OBJ-4** Estimate when the machine will need replacing.

MATERIALS & TOOLS

- ☐ A machine you own
- ☐ Cleaning supplies (Lab 10)
- ☐ Your machine record (Labs 1, 8)
- ☐ A calendar or reminder app

PROCEDURE

A Write the schedule

- A.1** Monthly: confirm backups completed, install updates, check free space (keep at least 15% free), and restart the machine fully.
- A.2** Quarterly: check drive SMART data, review startup items (Lab 16), run a malware scan, and check the battery report.
- A.3** Yearly: dust cleaning (twice a year in dusty homes or with pets), a test restore of the backup, a password audit, and a review of the machine record.

B Run it now

- B.1** Work through every item and tick it off. For drive health, compare today's SMART values with the ones from Lab 8.

WIN	Get-PhysicalDisk Get-StorageReliabilityCounter Select Wear, ReadErrorsTotal, Temperature
MAC	smartctl -a disk0
LINUX	sudo smartctl -a /dev/sda

- B.2** Log the date, what you found and what you did in the machine record.

B.3 Put reminders in your calendar for the next monthly, quarterly and yearly passes.

c Lifespan

C.1 Write down the machine's age, when OS support ends, battery health and drive wear. Estimate how many years it has left, and the trigger that will tell you to plan a replacement.

TROUBLESHOOTING

IF YOU SEE	TRY
Checklist too long to keep up with	Automate backups and updates, and keep only the items that need a human on the checklist.
SMART values have risen since the baseline	Check backups first, then plan to replace the drive. Don't wait for it to fail.
Free space keeps shrinking	Find what is growing: downloads, caches, old system restore points and cloud sync folders.

Building and Upgrading a Computer

Plan a fully compatible build or a specific upgrade, and justify every choice.

OBJECTIVES

- OBJ-1** Choose parts that fit each other on socket, memory type, power and physical size.
- OBJ-2** Write down your reason for each part.
- OBJ-3** Rank upgrades by benefit for the cost.
- OBJ-4** Plan the first boot and a no-POST recovery.

MATERIALS & TOOLS

- ☐ A budget and a clear purpose for the machine
- ☐ A PSU wattage calculator
- ☐ The manufacturer spec pages and the board's supported-CPU and supported-memory lists
- ☐ The machine you are upgrading (if upgrading)

PROCEDURE

A Define

- A.1** Write the purpose (office, photo editing, gaming, home server), the budget, and the one bottleneck you want gone (from Labs 6, 7, 8 and 18).

B Choose and check compatibility

- B.1** CPU and board: the sockets match, and the board's CPU support list includes your CPU and the firmware version it needs.
- B.2** Memory: the DDR generation matches the board, and the capacity and speed are on the board's supported list. Buy a matched kit.
- B.3** Storage: M.2 key and length, PCIe generation, and any SATA ports that are disabled when certain M.2 slots are used (see the board manual).
- B.4** Power and fit: $\text{total draw} \times 1.3 \leq \text{PSU rating}$. GPU length, cooler height and board size (ATX or mATX) all fit the case.
- B.5** For an upgrade instead of a build, confirm the exact slots and supported parts from the existing machine's manual and Lab 1 record.

C Plan the build day

- C.1** Write the assembly order: CPU, RAM and M.2 on the board outside the case, then a test boot on the box, then mount in the case, then cables, then the GPU.

CAUTION Follow the Lab 5 safety procedure. Never force the CPU into its socket or a connector into its port.

- C.2** Write the first boot plan: enter the firmware setup, check CPU, RAM and drives are detected, enable the memory profile (XMP or EXPO), install the OS from USB, then install drivers.

- C.3** Write the no-POST plan using your boot map: power, then the debug LED, then reseal RAM (one stick in A2), then the CPU power cable, then clear CMOS.

TROUBLESHOOTING

IF YOU SEE	TRY
The board doesn't support the new CPU out of the box	Update the firmware first with the old CPU, or use the board's BIOS Flashback if it has one.
RAM runs slower than rated	The memory profile is off. Enable XMP or EXPO in the firmware setup.
An upgrade costs more than half a new machine	Compare with a new machine. Older platforms also limit your future upgrades.
The M.2 drive isn't detected	Check the slot supports NVMe as well as SATA, and that the standoff screw is fitted.

Helping Others and Working in Support

Talk a real person through a real fix, using only questions and clear guidance.

OBJECTIVES

- OBJ-1** Get accurate symptoms from someone who isn't technical.
- OBJ-2** Guide a fix remotely without touching the machine.
- OBJ-3** Deliver an honest, kind summary, including bad news if needed.
- OBJ-4** Document the fix so it can be repeated.

MATERIALS & TOOLS

- ☐ A willing friend or family member with a real, minor problem
- ☐ A phone or video call
- ☐ Your troubleshooting log template (Lab 4)

PROCEDURE

A Listen first

- A.1** Ask open questions: What were you trying to do? What happened instead? When did it start? What changed? Write down their exact words.
- A.2** Play it back: "So when you click Print, nothing comes out, and it started after the new router?" Correct your understanding until they agree.

B Guide

- B.1** Ask them to describe what they see on screen before every instruction. Give one step at a time, using the words that actually appear on the screen.
- B.2** Before you ask them to change anything, say what you expect to happen. Undo anything that doesn't help.
- B.3** If screen sharing is an option, use only a trusted built-in tool, with their consent, and end the session when you are done.

WIN	Quick Assist
MAC	Screen Sharing via FaceTime or Messages
LINUX	GNOME Remote Desktop / RustDesk

C Close and document

- C.1** Explain the cause and the fix in two sentences with no jargon. If the news is bad (for example, a dying drive), say what it means for their data and what to do next.
- C.2** Log the problem, the questions that cracked it, the fix, and what you would do faster next time.

TROUBLESHOOTING

IF YOU SEE	TRY
They say "nothing happens"	Ask what is on the screen right now, and ask them to read it out word for word.
They are frustrated or embarrassed	Say the problem is common and not their fault. Then get to work.
The fix is beyond remote help	Say so early and arrange to see the machine or send it for repair. Make sure it is backed up first.
They ask you to handle their passwords	Have them type passwords themselves. Never store another person's credentials.

Keeping Up and Where to Go Next

Rate your skills against the whole course and write your 90-day plan.

OBJECTIVES

- OBJ-1** Rate your confidence against every lab.
- OBJ-2** Pick a direction: hardware, networking, operating systems, security or programming.
- OBJ-3** Write a specific 90-day plan with weekly practice.

MATERIALS & TOOLS

- ☐ This guide and your notes
- ☐ A calendar
- ☐ Any old machines available for a home lab

PROCEDURE

A Self-assessment

- A.1** List Labs 1–31 and rate each 1–3: 1 = I could follow it again, 2 = I could do it without the guide, 3 = I could teach it.
- A.2** Circle every lab you rated 1. Redo the three most important of them this month.

B Choose a direction

- B.1** Write which part of the course you enjoyed most: the physical hardware (Part II), the software (Part III), networks and security (Part IV), or diagnosis (Part V). This points you toward a direction.
- B.2** For your direction, list one entry-level certification, one community you could join, and one project for a home lab. Look up current certification objectives; they change.

C The 90-day plan

- C.1** Days 1–30: consolidate. Redo the labs you circled and start a home lab machine.
- C.2** Days 31–60: go deeper. Take one course or study guide in your direction and do weekly practice on the home lab.
- C.3** Days 61–90: prove it. Help three people (Lab 31), document every fix, and book or pass a certification exam if you chose one.

C.4 Put the plan in your calendar with one fixed practice session each week.

TROUBLESHOOTING

IF YOU SEE

TRY

Everything feels equally interesting

Pick the direction with the closest practical project, such as building a home network or a media server. Doing the project will show you what you enjoy.

No old hardware for a home lab

Use virtual machines (VirtualBox, UTM, virt-manager) on your current computer.

You lose momentum after two weeks

Make it smaller: 30 minutes a week, the same day every week, with a lab log entry each time.
